

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
Electronics and Communication Engineering
Basic Electronics (24B11EC111)
Tutorial No. 8

- Q1.** Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{\text{ox}} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V}\cdot\text{s}$, and $V_t = 0.7 \text{ V}$.
[CO4]
- (a) Find C_{ox} and k_n' .
- (b) For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
- (c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a 1000Ω resistor for very small v_{DS} .
- [Ans. $4.32 \times 10^{-3} \text{ F/m}^2$, $194 \mu\text{A/V}^2$, 0.32 V , 1.02 V , 0.32 V , 0.52 V , 1.22 V]**
- Q2.** A circuit designer intending to operate a MOSFET in saturation is considering the effect of changing the device dimensions and operating voltages on the drain current I_D . Specifically, by what factor does I_D change in each of the following cases?
[CO4]
- (a) The channel length is doubled.
- (b) The channel width is doubled.
- (c) The overdrive voltage is doubled.
- (d) The drain-to-source voltage is doubled.
- (e) Changes (a), (b), (c), and (d) are made simultaneously.
- Which of these cases might cause the MOSFET to leave the saturation region?
- [Ans. 0.5; 2; 4; no change; 4; case (c) if v_{DS} is smaller than $2V_{\text{OV}}$]**
- Q3.** Consider an NMOS transistor fabricated in a $0.18\text{-}\mu\text{m}$ process with $L = 0.18 \mu\text{m}$ and $W = 2 \mu\text{m}$. The process technology is specified to have $C_{\text{ox}} = 8.6 \text{ fF}/\mu\text{m}^2$, $\mu_n = 450 \text{ cm}^2/\text{V}\cdot\text{s}$, and $V_{\text{tn}} = 0.5 \text{ V}$.
[CO4]
- (a) Find V_{GS} and V_{DS} that result in the MOSFET operating at the edge of saturation with $I_D = 100 \mu\text{A}$.
- (b) If V_{GS} is kept constant, find V_{DS} that results in $I_D = 50 \mu\text{A}$.
- [Ans. 0.72 V , 0.22 V , 0.06 V]**
- Q4.** Consider NMOS as shown in *Fig. 1*, what will be the change in I_D if V_{DD} changes from 3.3 V to 1.8 V . Consider $V_{\text{IN}} = 2.5 \text{ V}$, $V_{\text{TH}} = 0.5 \text{ V}$ and $\mu_n C_{\text{ox}} W/L = 100 \mu\text{A/V}^2$?
[CO4]

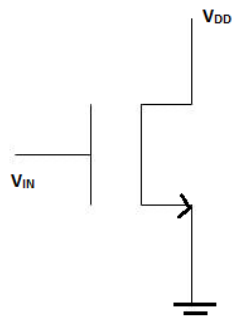


Figure 1

[Ans. 2 μA]